

Optimal Control

P. 18,19 - Robot Optimization

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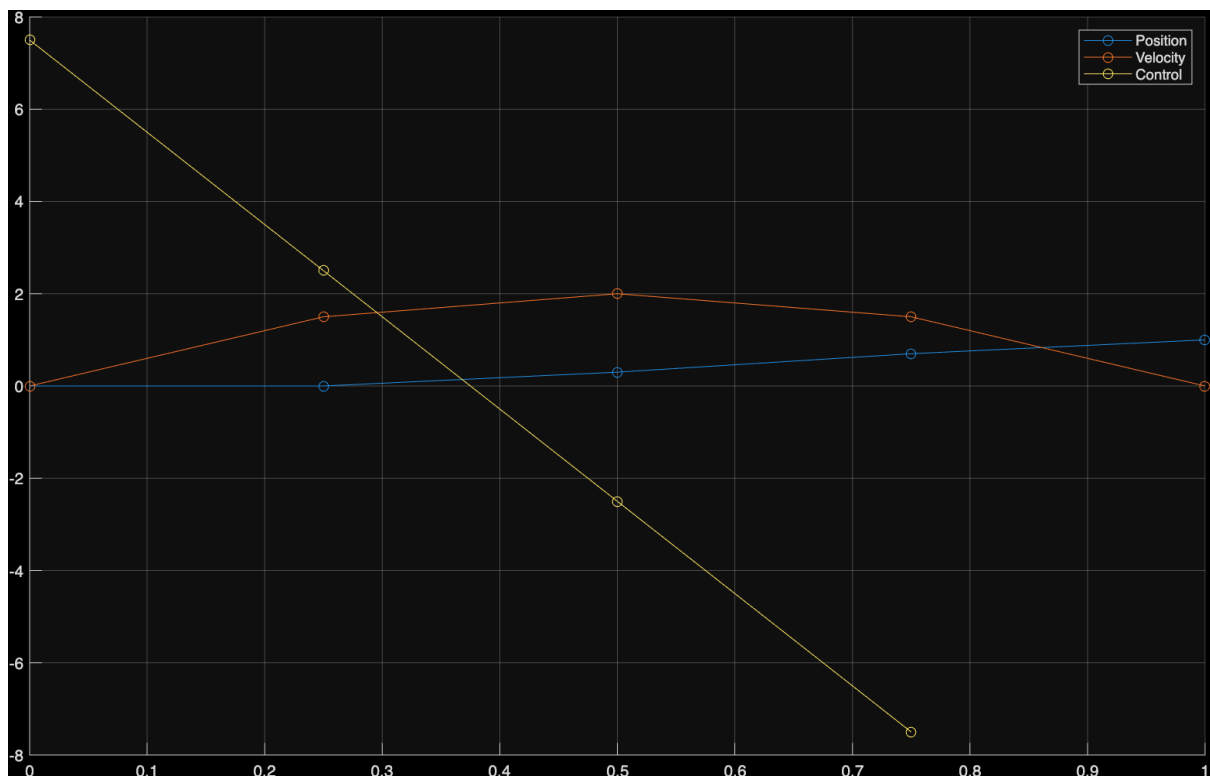
1. Problem 18

Solve the optimal control problem $\ddot{x} = u$ subject to $x(0) = 0, x(1) = 1, v(0) = v(1) = 0$

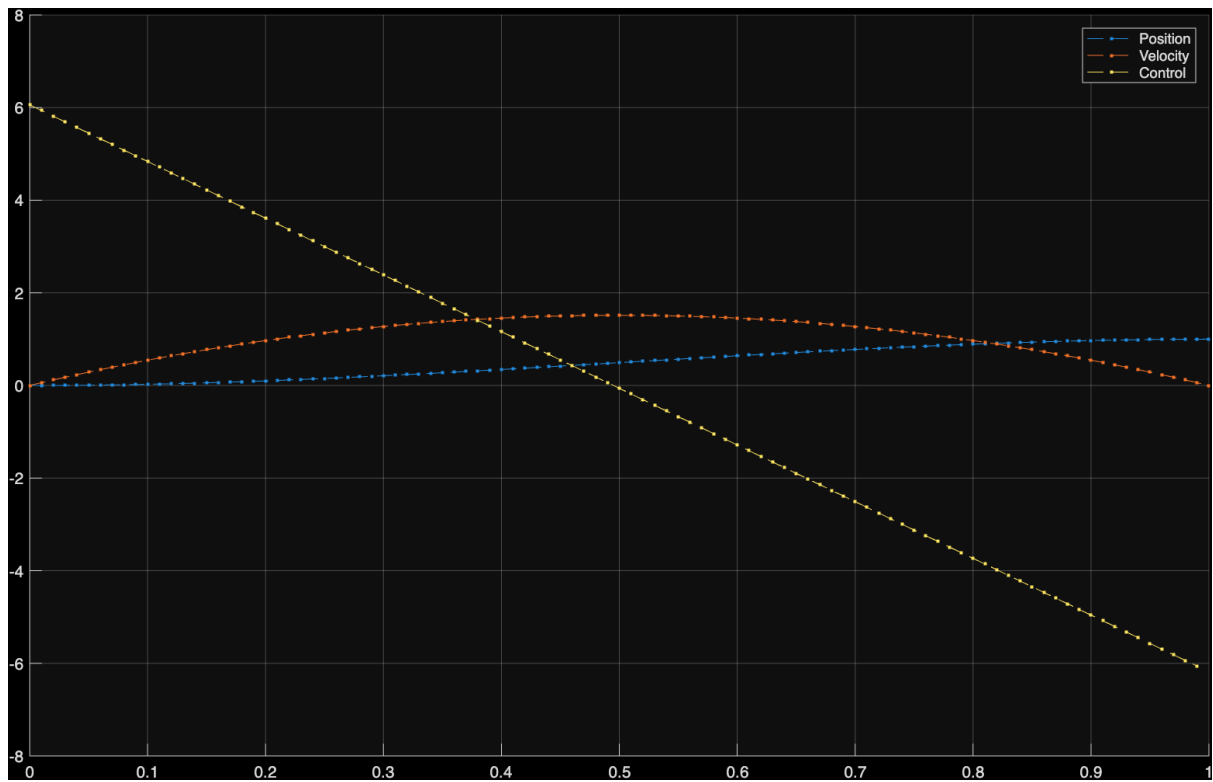
My $z = [x_0, x_1, \dots, x_N, v_0, v_1, \dots, v_N, u_0, u_1, \dots, u_{N-1}]$

So number of x equations is $N+1$, number of v equations is $N+1$, number of u equations is N .

1.1. $N = 4$



1.2. $N = 100$



1.3. Code - Using nonlcon

```
clear; close all; clc;

N = 100;
t0 = 0;
tf = 1;
h = (tf-t0) / (N + 1);

z_guess = zeros(2*(N+1) + N, 1);
cost_func = @(z) obj_fun(z, N, h);
nonlcon = @(z) const_fun(z, N, h);

options = optimoptions('fmincon', 'Display', 'iter',
    'Algorithm', 'sqp');
[z_opt, fval] = fmincon(cost_func, z_guess, ...
    [], [], [], [], [], ...
    nonlcon, ...
    options);

% Unpack solution
x1 = z_opt(1 : N+1);
x2 = z_opt(N+2 : 2*N+2);
u = z_opt(2*N+3 : end);
t = linspace(t0, tf, N+1);

figure()
hold on;
plot(t, x1, '--.', DisplayName='Position');
plot(t, x2, '--.', DisplayName='Velocity');
plot(t(1: end-1), u, '--.', DisplayName='Control');
legend();
grid on;

function J = obj_fun(z, N, h)
    u = z(2*N+3 : end);
    J = sum(u(1:N).^2) * h;
end
```

```

function [c, ceq] = const_fun(z, N, h)
    x1 = z(1 : N+1);
    x2 = z(N+2 : 2*N+2);
    u   = z(2*N+3 : end);

    ceq = [];

    % Boundary Constraints
    ceq = [ceq; x1(1) - 0];
    ceq = [ceq; x2(1) - 0];
    ceq = [ceq; x1(N+1) - 1];
    ceq = [ceq; x2(N+1) - 0];

    % Dynamics
    for i = 1:N
        ceq = [ceq; x1(i+1) - (x1(i) + h * x2(i))];
        ceq = [ceq; x2(i+1) - (x2(i) + h * u(i))];
    end

    c = []; % No inequality constraints
end

```

2. Problem 19 - Trapezoidal Discretization

Solve above problem using trap. disc. and compare accuracy with Euler solution.